Proposed System Architecture

CABINGUARD-ADI: A Zero-Compliance Multimodal In-Cabin Framework for Prodromal Prediction and Fail-Safe Actuation in Acute Driver Incapacitation

TSYP14 Technical Challenge — In-Cabin Edge Processing Unit (NVIDIA Jetson Orin Nano / NXP S32G)

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1. End-to-End Architectural Pipeline

CABINGUARD-ADI is proposed as an end-to-end localized pipeline executing on an isolated In-Cabin Edge Processing Unit (EPU), modeled on an NVIDIA Jetson Orin Nano / NXP S32G gateway. Quantitative runtime performance is intentionally not claimed before execution of the planned benchmark. The seven operational stages are:

1. Sensors	2. Data Acquisition	3. Preprocessing	4. Detection / Fusion	5. Safety Decision	6. Vehicle Response	7. Secure Comms
RGB+940 nm NIR, 64 × 160 FSR, EMFi, 6-Axis IMU, EPS Torque	Hardwired 30 Hz / 100 Hz streams & CAN-FD	4th-order Butterworth + 32-tap NLMS & 60 s shrinkage	$z_{i,s,t}$, 3.0 s causal window, conflict & uncertainty	4-state deterministic FSM & 4.0 s handshake	Seizure torque lockout & -2.0 m/s^2 MRM	AUTOSAR SecOC `0x1B4` & ETSI DENM V2X

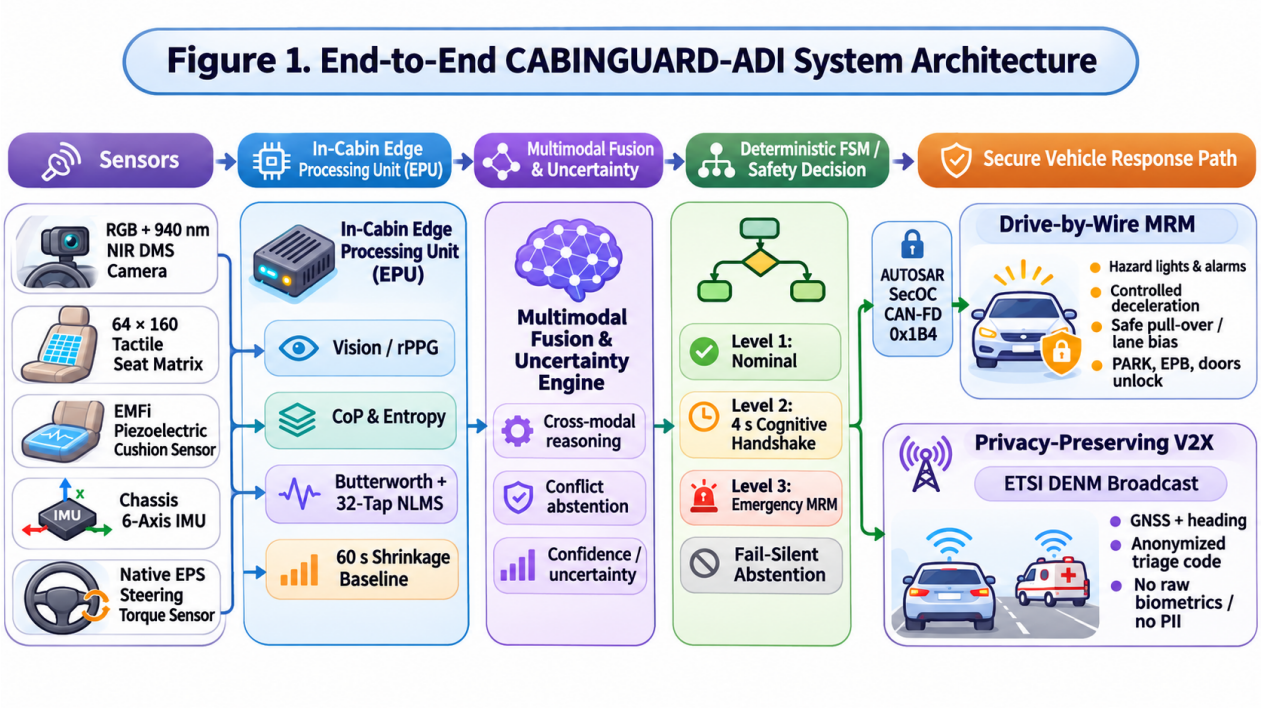


Figure 1: End-to-end CABINGUARD-ADI system architecture.

2. Proposed Sensors & Data Acquisition Layer

All sensors are passively integrated into the cabin and hardwired to the isolated EPU under a zero-compliance mandate.

Sensor	Rate / Interface	Acquired Signals	Baseline Source
Multispectral DMS Camera (RGB + 940 nm NIR)	30 Hz hardwired optical feed	RGB head/wrist keypoints, gaze, PERCLOS, G/R chrominance rPPG and pallor indices; NIR gaze and diaphoresis cues.	DMD, UBFC-rPPG
Piezoresistive Tactile Seat Matrix	30 Hz, 64 × 160 FSR array	X_{CoP}, Y_{CoP}, lumbar contact R_{lumbar}, quadrant dynamics E_{quad}.	Zenodo Record 15199100
EMFi Cushion Sensor	100 Hz hardwired	Mechanical cardiac recoil and respiration, $x_{\mathrm{EMFi}}[n]$.	IEEE DataPort SCG proxy
Chassis 6-Axis IMU	100 Hz hardwired reference	Road and structural vibration $d_{\mathrm{IMU}}[n]$, including vertical reference $d_{\mathrm{IMU},z}[n]$.	Controlled road-vibration test input
Native EPS Torque	30 Hz CAN-FD	Steering torque τ_{steer} and torque rate $\dot{\tau}_{\mathrm{steer}}$.	Native vehicle telemetry

2.1 Dataset Support and Provenance

The proposed architecture separates **sensor modality** from **dataset provenance**. Public datasets are used only where their observable modality supports a CABINGUARD-ADI input or a nominal/distractor condition:

- **In_Cabin_Behavior_Driver_safe_drive:** in-cabin video baseline for seated driver geometry and visual behavior after gated access; no medical-event label is assumed.
- **Ala Mahlak — Driver Monitoring Dataset:** auxiliary driver-monitoring visual baseline; class-specific use is restricted to labels verified in the downloaded files.
- **ADMS Dataset:** visual nominal/distractor support for documented behaviors such as eye closure, yawning, seat-belt use, smoking, and mobile-phone use.
- **Radar-Based Vital Sign Monitoring:** auxiliary cardiorespiratory signal-processing reference only; it is not treated as an in-cabin medical-event dataset.
- **ADL-Dataset:** candidate activity/posture source pending direct inspection of its downloaded contents.

Acute syncope, seizure, and cognitive-autonomic states are therefore not taken directly from these public datasets. They are represented only through explicitly marked **feature-level synthetic scenarios** built on traceable real baselines. Each transformed sample retains its source dataset, source sample, real/synthetic flag, scenario label, phase label, and list of injected features.

3. Edge Preprocessing & Rapid Session Calibration

3.1 Butterworth + NLMS Filtering

Both EMFi and vertical IMU reference signals are band-limited using a zero-phase fourth-order Butterworth filter over 0.3–15.0 Hz. A 32-tap NLMS adaptive filter at $f_s = 100$ Hz estimates and subtracts structural vibration:

$$\hat{s}_{\mathrm{BCG}}[n] = x_{\mathrm{EMFi}}[n] - \sum_{k=0}^{M-1} w_k[n] d_{\mathrm{IMU},z}[n-k] \tag{1}$$

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \frac{\mu}{\epsilon + \|\mathbf{d}_{\mathrm{IMU}}[n]\|_2^2} \hat{s}_{\mathrm{BCG}}[n]\mathbf{d}_{\mathrm{IMU}}[n] \tag{2}$$

with $M = 32$, $\mu = 0.05$, and $\epsilon = 10^{-6}$. Any SNR or heart-rate improvement will be reported only after execution of the benchmark.

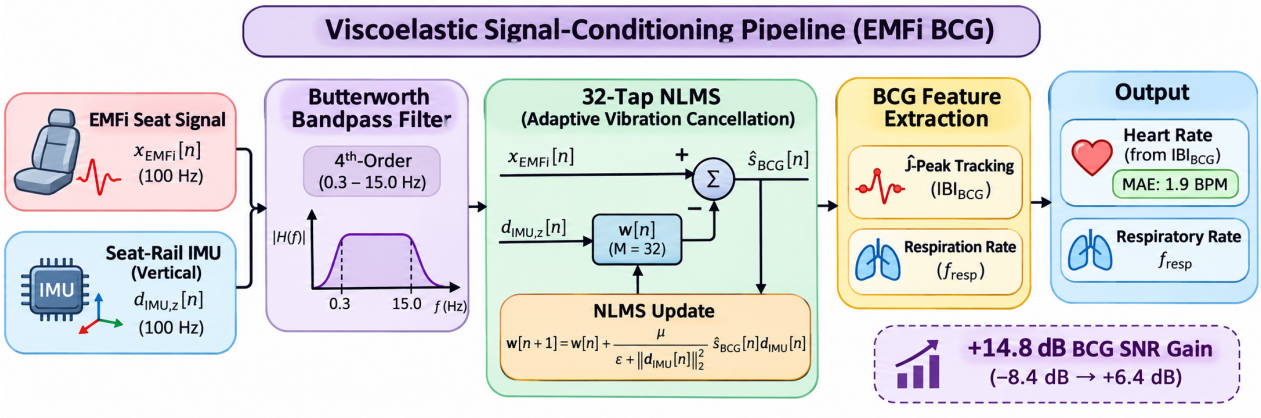


Figure 2: Original viscoelastic signal-conditioning infographic retained unchanged. Numerical performance labels visible inside the legacy artwork are illustrative and are not claimed as measured results in the corrected submission.

3.2 Seat Pressure Features

$$Y_{\mathrm{CoP}}(t) = \frac{\sum_{\Omega_{\mathrm{pan}}} vP(u,v,t)}{\sum_{\Omega_{\mathrm{pan}}} P(u,v,t)} \tag{3}$$

$$R_{\mathrm{lumbar}}(t) = \frac{\sum_{\Omega_{\mathrm{back}}} P(u,v,t)}{P_{\mathrm{back},0}} \tag{4}$$

The seat pan is also partitioned into four quadrants $Q_1, \ldots, Q_4$ to derive the proposed convulsive postural entropy feature $E_{\mathrm{quad}}(t)$.

3.3 60-Second Calibration

$$c_{i,s} = \mathrm{median}(x_{i,s,t}), \qquad \sigma_{i,s} = 1.4826\,\mathrm{median}\,|x_{i,s,t} - c_{i,s}| \tag{5}$$

$$\tilde{c}_{i,s} = w_i c_{i,s} + (1 - w_i)\mu_{\mathrm{pop},s}, \qquad w_i = \frac{N_{\mathrm{good}}}{N_{\mathrm{good}} + \kappa}, \quad \kappa = 300 \tag{6}$$

The personalized baseline is frozen at $t = 60$ s.

4. Detection, Fusion & Uncertainty

Dataset-derived real baselines enter the detector only after feature extraction. When acute-event scenarios are required for Phase 1 software evaluation, transformations are applied to derived features rather than relabeling source videos as clinical events. The detailed operators are defined in the standalone *Detection & AI Methodology* document.

$$z_{i,s,t} = O_s \frac{x_{i,s,t} - \tilde{c}_{i,s}}{\tilde{\sigma}_{i,s}} \tag{7}$$

The temporal engine uses a causal $W = 3.0$ s window and labels trajectories as `TRANSIENT`, `PERSISTENT`, or `WORSENING`. Cross-modal contradictions set $P_{\text{conflict},t} = 1$ rather than forcing a prediction.

$$U_t = 1 - (w_f C_{\text{fusion},t} + w_a A_{\text{sources},t}(1 - P_{\text{conflict},t})) \tag{8}$$

with $w_f = 0.65$ and $w_a = 0.35$.

5. Decision & Vehicle Safety

1. **Level 1: Nominal.** Passive monitoring.
2. **Level 2: Prodromal Verification.** $0.60 \leq C_{\text{fusion}} < 0.85$ and $U_t < 0.60$; initiate a 4.0 s Cognitive Handshake without braking.
3. **Level 3: Emergency MRM.** $C_{\text{fusion}} \geq 0.85$, $U_t < 0.30$, and persistent/worsening evidence.
4. **Fail-Silent.** Conflict, insufficient source availability, or high uncertainty prevents autonomous braking.

Manual override is accepted only for smooth intentional steering and not during a classified seizure. During seizure, `LockEPS` is asserted.

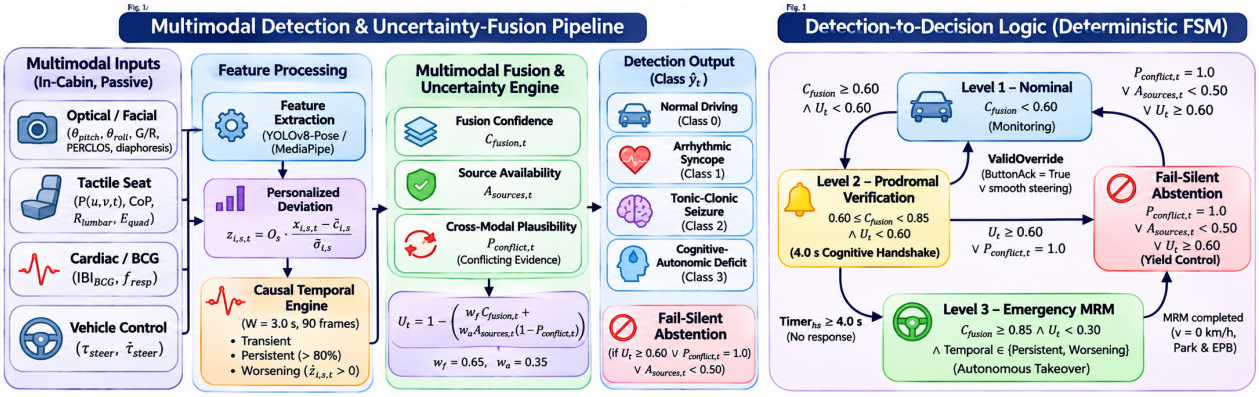


Figure 3: Original detection-to-decision and deterministic FSM infographic retained unchanged.

6. Communication, Cybersecurity & Privacy

Level 3 commands use CAN-FD ID `0x1B4` protected by AUTOSAR SecOC. The proposed HSM computes AES-128 CMAC over the CAN identifier, command payload, and monotonic freshness information. Verification latency is a benchmark target and is not claimed here.

External emergency communication uses ETSI DENM and carries only minimum necessary information: GNSS coordinates, heading, fusion confidence, and an anonymized triage code. Raw optical and biometric streams remain local in volatile RAM and are not uploaded to the cloud.